

```
pip install trl==0.11.3 fsspec==2025.3.2 --upgrade
```

```

Downloading nvitoolkit-12.5.82-cuda12.5-linux-x86_64-gnu-glibc-ubuntu20.04_1.whl (207.5 MB)
207.5/207.5 MB 5.5 MB/s eta 0:00:00
Downloading nvjitlink-cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (21.1 MB)
21.1/21.1 MB 105.8 MB/s eta 0:00:00
Downloading tyro-0.9.19-py3-none-any.whl (124 kB)
124.3/124.3 kB 12.4 MB/s eta 0:00:00
Downloading datasets-2.14.4-py3-none-any.whl (519 kB)
519.3/519.3 kB 42.8 MB/s eta 0:00:00
Downloading dill-0.3.7-py3-none-any.whl (115 kB)
115.3/115.3 kB 11.9 MB/s eta 0:00:00
Downloading shtab-1.7.2-py3-none-any.whl (14 kB)
Downloading multiprocessing-0.70.15-py311-none-any.whl (135 kB)
135.4/135.4 kB 14.4 MB/s eta 0:00:00
Downloading xxhash-3.5.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (194 kB)
194.8/194.8 kB 19.3 MB/s eta 0:00:00
Installing collected packages: xxhash, shtab, nvidia-nvjitlink-cu12, nvidia-cufft-cu12, nvidia-cuda-runtime-cu12,
Attempting uninstall: nvidia-nvjitlink-cu12
Found existing installation: nvidia-nvjitlink-cu12 12.5.82
Uninstalling nvidia-nvjitlink-cu12-12.5.82:
Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82
Attempting uninstall: nvidia-curand-cu12
Found existing installation: nvidia-curand-cu12 10.3.6.82
Uninstalling nvidia-curand-cu12-10.3.6.82:
Successfully uninstalled nvidia-curand-cu12-10.3.6.82
Attempting uninstall: nvidia-cufft-cu12
Found existing installation: nvidia-cufft-cu12 11.2.3.61
Uninstalling nvidia-cufft-cu12-11.2.3.61:
Successfully uninstalled nvidia-cufft-cu12-11.2.3.61
Attempting uninstall: nvidia-cuda-runtime-cu12
Found existing installation: nvidia-cuda-runtime-cu12 12.5.82
Uninstalling nvidia-cuda-runtime-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82
Attempting uninstall: nvidia-cuda-nvrtc-cu12
Found existing installation: nvidia-cuda-nvrtc-cu12 12.5.82
Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82
Attempting uninstall: nvidia-cuda-cupti-cu12
Found existing installation: nvidia-cuda-cupti-cu12 12.5.82
Uninstalling nvidia-cuda-cupti-cu12-12.5.82:
Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82
Attempting uninstall: nvidia-cublas-cu12
Found existing installation: nvidia-cublas-cu12 12.5.3.2
Uninstalling nvidia-cublas-cu12-12.5.3.2:
Successfully uninstalled nvidia-cublas-cu12-12.5.3.2
Attempting uninstall: nvidia-cusparse-cu12
Found existing installation: nvidia-cusparse-cu12 12.5.1.3
Uninstalling nvidia-cusparse-cu12-12.5.1.3:
Successfully uninstalled nvidia-cusparse-cu12-12.5.1.3
Attempting uninstall: nvidia-cudnn-cu12
Found existing installation: nvidia-cudnn-cu12 9.3.0.75
Uninstalling nvidia-cudnn-cu12-9.3.0.75:
Successfully uninstalled nvidia-cudnn-cu12-9.3.0.75
Attempting uninstall: nvidia-cusolver-cu12
Found existing installation: nvidia-cusolver-cu12 11.6.3.83
Uninstalling nvidia-cusolver-cu12-11.6.3.83:
Successfully uninstalled nvidia-cusolver-cu12-11.6.3.83
Successfully installed datasets-2.14.4 dill-0.3.7 multiprocessing-0.70.15 nvidia-cublas-cu12-12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nv

```

```

import torch
from tqdm import tqdm
import pandas as pd

tqdm.pandas()

import datasets

from transformers import pipeline, AutoTokenizer
from datasets import load_dataset

from trl import PPOTrainer, PPOConfig, AutoModelForCausalLMWithValueHead
from trl.core import LengthSampler

# First upgrade your datasets library (recommended)
!pip install -U datasets transformers

from datasets import load_dataset, disable_caching
from transformers import AutoTokenizer

```

```
from torch.utils.data import DataLoader
import torch
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.11/dist-packages (3.5.1)
Requirement already satisfied: transformers in /usr/local/lib/python3.11/dist-packages (4.51.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from datasets) (3.18.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.3.7)
Requirement already satisfied: pandas in /usr/local/lib/python3.11/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.11/dist-packages (from datasets) (2.32.3)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.11/dist-packages (from datasets) (4.67.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.11/dist-packages (from datasets) (3.5.0)
Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.70.15)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0) (2025.3.0)
Requirement already satisfied: aiohttp in /usr/local/lib/python3.11/dist-packages (from datasets) (3.11.15)
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.30.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from datasets) (24.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages (from datasets) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist-packages (from transformers) (2024.11.6)
Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.21.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.5.3)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->datasets) (2.6.1)
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from aiohttp->datasets) (1.3.2)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->datasets) (25.3.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp->datasets) (1.6.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp->datasets) (6.4.3)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->datasets) (0.3.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp->datasets) (1.20.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.24.0->data) (4.12.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2.4.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2025.1.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
```

```
print(datasets.__version__)
```

```
3.5.1
```

```
torch.cuda.empty_cache()
```

```
config = PPOConfig(
    model_name="distilbert/distilgpt2",
    learning_rate=1.41e-5,
)
```

```
sent_kwargs = {"return_all_scores": True, "function_to_apply": "none", "batch_size": 8}
```

```
/usr/local/lib/python3.11/dist-packages/trl/trainer/ppo_config.py:207: FutureWarning: `PPOConfig` is deprecated and will be removed in trl v0.10.0. Please use `PPOConfig` instead.
warnings.warn(
```

```
def build_dataset(config, dataset_name="Lucylulu/imdb", revision="main", input_min_text_length=4, input_max_text_length=12):
    """
```

```
    Build dataset for training. This builds the dataset from `load_dataset`, one should
    customize this function to train the model on its own dataset.
```

```
Args:
```

```
    dataset_name (`str`):
        The name of the dataset to be loaded.
```

```
Returns:
```

```
    dataloader (`torch.utils.data.DataLoader`):
        The dataloader for the dataset.
```

```
"""
```

```
tokenizer = AutoTokenizer.from_pretrained(config.model_name)
tokenizer.pad_token = tokenizer.eos_token
# load imdb with datasets
ds = load_dataset(dataset_name, split="train", revision=revision)
```

```
ds = ds.filter(lambda x: (len(x["text"]) > 100 if x["text"] is not None else False), batched=False)
```

```

input_size = LengthSampler(input_min_text_length, input_max_text_length)

def tokenize(sample):
    sample["input_ids"] = tokenizer.encode(sample["text"])[0: input_size()]
    sample["query"] = tokenizer.decode(sample["input_ids"])
    return sample

ds = ds.map(tokenize, batched=False)
ds.set_format(type="torch")
return ds

dataset = build_dataset(config)

def collator(data):
    return dict((key, [d[key] for d in data]) for key in data[0])

```

 train.jsonl: 100% 26.9M/26.9M [00:00<00:00, 70.2MB/s]
 dev.jsonl: 100% 6.81M/6.81M [00:00<00:00, 17.8MB/s]
 test.jsonl: 100% 33.0M/33.0M [00:00<00:00, 69.8MB/s]
 Generating train split: 100% 20000/20000 [00:00<00:00, 100910.85 examples/s]
 Generating validation split: 100% 5000/5000 [00:00<00:00, 90895.98 examples/s]
 Generating test split: 100% 25000/25000 [00:00<00:00, 94679.50 examples/s]
 Filter: 100% 20000/20000 [00:00<00:00, 167563.38 examples/s]
 Map: 100% 19993/19993 [00:24<00:00, 918.39 examples/s]
 Token indices sequence length is longer than the specified maximum sequence length for this model (1053 > 1024). Running this sequence t



```

print(dataset)

```

 Dataset({
 features: ['id', 'text', 'label', 'input_ids', 'query'],
 num_rows: 19993
})

```

model = AutoModelForCausalLMWithValueHead.from_pretrained(config.model_name)
ref_model = AutoModelForCausalLMWithValueHead.from_pretrained(config.model_name)
tokenizer = AutoTokenizer.from_pretrained(config.model_name)

tokenizer.pad_token = tokenizer.eos_token

```

 Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to regular HTTP download. For better perfc
 WARNING:huggingface_hub.file_download:Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to r
 model.safetensors: 100% 353M/353M [00:01<00:00, 226MB/s]
 generation_config.json: 100% 124/124 [00:00<00:00, 7.97kB/s]
 Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to regular HTTP download. For better perfc
 WARNING:huggingface_hub.file_download:Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to r
 pytorch_model.bin: 100% 353M/353M [00:01<00:00, 247MB/s]



```

ppo_trainer = PPOTrainer(config, model, ref_model, tokenizer, dataset=dataset, data_collator=collator)

```

 /usr/local/lib/python3.11/dist-packages/trl/trainer/ppo_trainer.py:193: FutureWarning: `PPOTrainer` is deprecated and will be removed in
 warnings.warn(



```

device = ppo_trainer.accelerator.device
if ppo_trainer.accelerator.num_processes == 1:
    device = 0 if torch.cuda.is_available() else "cpu" # to avoid a `pipeline` bug
sentiment_pipe = pipeline("sentiment-analysis", model="ealvaradob/bert-finetuned-phishing", device=device)

```

```

config.json: 100% 845/845 [00:00<00:00, 54.6kB/s]
pytorch_model.bin: 100% 1.34G/1.34G [00:11<00:00, 102MB/s]
model.safetensors: 100% 1.34G/1.34G [00:22<00:00, 19.1MB/s]
tokenizer_config.json: 100% 1.19k/1.19k [00:00<00:00, 16.8kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 3.43MB/s]
tokenizer.json: 100% 711k/711k [00:00<00:00, 9.54MB/s]
special_tokens_map.json: 100% 125/125 [00:00<00:00, 3.43kB/s]

```

```

text = "This movie is astonishingly poor"
sentiment_pipe(text, **sent_kwargs)

```

```

/usr/local/lib/python3.11/dist-packages/transformers/pipelines/text_classification.py:106: UserWarning: `return_all_scores` is now deprecated
warnings.warn(
[[{'label': 'benign', 'score': 4.920907974243164},
{'label': 'phishing', 'score': -3.4827210903167725}]]

```

```

gen_kwargs = {"min_length": -1, "top_k": 0.0, "top_p": 1.0, "do_sample": True, "pad_token_id": tokenizer.eos_token_id}

```

```

output_min_length = 6
output_max_length = 20
output_length_sampler = LengthSampler(output_min_length, output_max_length)

```

```

generation_kwargs = {
    "min_length": -1,
    "top_k": 0.0,
    "top_p": 1.0,
    "do_sample": True,
    "pad_token_id": tokenizer.eos_token_id,
}

```

```

for epoch, batch in tqdm(enumerate(ppo_trainer.dataloader)):
    query_tensors = batch["input_ids"]

    #### Get response from gpt2
    response_tensors = []
    for query in query_tensors:
        gen_len = output_length_sampler()
        generation_kwargs["max_new_tokens"] = gen_len
        response = ppo_trainer.generate(query, **generation_kwargs)
        response_tensors.append(response.squeeze()[-gen_len:])
    batch["response"] = [tokenizer.decode(r.squeeze()) for r in response_tensors]

    #### Compute sentiment score
    texts = [q + r for q, r in zip(batch["query"], batch["response"])]
    pipe_outputs = sentiment_pipe(texts, **sent_kwargs)
    rewards = [torch.tensor(output[1]["score"]) for output in pipe_outputs]

    #### Run PPO step
    stats = ppo_trainer.step(query_tensors, response_tensors, rewards)
    ppo_trainer.log_stats(stats, batch, rewards)

```

```

0it [00:00, ?it/s]/usr/local/lib/python3.11/dist-packages/transformers/pipelines/text_classification.py:106: UserWarning: `return_all_scores` is now deprecated
warnings.warn(
156it [35:10, 13.53s/it]

```

```

#### get a batch from the dataset
bs = 16
game_data = dict()
dataset.set_format("pandas")
df_batch = dataset[:].sample(bs)
game_data["query"] = df_batch["query"].tolist()
query_tensors = df_batch["input_ids"].tolist()

```

```
response_tensors_ref, response_tensors = [], []
```

```
#### get response from gpt2 and gpt2_ref
```

```
for i in range(bs):
```

```
    gen_len = output_length_sampler()
```

```
    output = ref_model.generate(
```

```
        torch.tensor(query_tensors[i]).unsqueeze(dim=0).to(device), max_new_tokens=gen_len, **gen_kwargs
```

```
    ).squeeze()[:-gen_len:]
```

```
    response_tensors_ref.append(output)
```

```
    output = model.generate(
```

```
        torch.tensor(query_tensors[i]).unsqueeze(dim=0).to(device), max_new_tokens=gen_len, **gen_kwargs
```

```
    ).squeeze()[:-gen_len:]
```

```
    response_tensors.append(output)
```

```
#### decode responses
```

```
game_data["response (before)"] = [tokenizer.decode(response_tensors_ref[i]) for i in range(bs)]
```

```
game_data["response (after)"] = [tokenizer.decode(response_tensors[i]) for i in range(bs)]
```

```
#### sentiment analysis of query/response pairs before/after
```

```
texts = [q + r for q, r in zip(game_data["query"], game_data["response (before)"])]
```

```
game_data["rewards (before)"] = [output[1]["score"] for output in sentiment_pipe(texts, **sent_kwargs)]
```

```
texts = [q + r for q, r in zip(game_data["query"], game_data["response (after)"])]
```

```
game_data["rewards (after)"] = [output[1]["score"] for output in sentiment_pipe(texts, **sent_kwargs)]
```

```
# store results in a dataframe
```

```
df_results = pd.DataFrame(game_data)
```

```
df_results
```

 /usr/local/lib/python3.11/dist-packages/transformers/pipelines/text_classification.py:106: UserWarning: `return\_all\_scores` is now deprecated. Use `return\_token\_embeddings` instead.
warnings.warn(

	query	response (before)	response (after)	rewards (before)	rewards (after)
0	The premise of the movie has been explained an...	are a director of some kind (but if	mistake this, you go a five or ten	-5.700898	-3.867198
1	Based on an actual mining disaster	– we didn't have to rely solely on correlatio...	mining disaster that can inflict immediate ec...	-5.352339	6.176741
2	I haven't seen it in	a while as at one point, no matter how much G...	haven't seen it in over sixteen years! Subscr...	-5.741900	5.423460
3	Everyone does things that they later regret	or trouble themselves."	! ** Click play here.	-4.516530	6.069374
4	My 2 year old likes the Doodlebops show	Hew T-shirts and updates on Falling Star from...	2 year old likes the Doodlebops show you's ex...	-4.546093	6.096477
5	Invasion of the Star Creatures	, Flying Vanquished Diodorus, the Kraken,	! A Deathsuper Health Miner Imperial Battle Ba...	2.457457	5.795339
6	I chose this movie by	Mine. The film opens and the backend begins:\...	chose this movie by O! Fantasy Sign Now! Sign...	-5.671598	6.090403
7	IMDb lists this as 1972 for some	of the most popular WWII Photo Wallpapers. It...	of the finest in the world. Smotreme Video\n\...	-5.420957	-4.819845
8	I don't give much	thought obviously to Southeast Asians and I d...	value here, send it! Buy it 360Weekulate Options	-5.627118	6.115571
9	I do miss the company Vestron, they sure	are are best known about the sentimental Japa...	miss the company Vestron, they sure will USD ...	-5.463335	4.083571
10	Advertised by channel seven in	Advertised by channel seven in 4< endoftext >	4 top pictures. Click here for billion miles ...	3.395850	6.129587
11	Great acting, great movie. If you are	going to your local comedy bar, here	an experienced film. Live in action.	4.699718	4.253224
12	It's very sly for all of the 60's look	ups and everybody showed up. Seriously, they	for all of the 60's look!< endoftext >	-4.879254	4.700229
13	Military training films are becoming so common...	appear at least occasionally in a backpacking...	can be bought right here. Download a 62-song ...	-4.612953	5.283574
14	Just okay film about a woman who	receives HIV after she is separated. i got al...	risks the world with her trust? Download the ...	-4.983489	6.070128
15	I thought it was not the best	idea, but the one I noticed was in	best proposal. Download it here now! < endofte...	-5.521297	5.739458

Next steps: [Generate code with df_results](#) [View recommended plots](#) [New interactive sheet](#)

